

Shalini Cherukuri

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EDUCATION

MS Computer science, *Indiana University Bloomington*
CGPA : 3.9/4.0

08/2023 – 05/2025 | Bloomington, IN

BE Computer Science and Engineering, *PSG College of Technology*

08/2018 – 05/2022 | Coimbatore, India

PROFESSIONAL EXPERIENCE

Senior Consultant (SDE), *Heartland Community Network*

06/2025 – Present | Remote, US

- Spearheaded development of full-stack web applications using the **MERN** stack leveraging **MongoDB** for data storage, **Express.js** and **Node.js** for backend APIs, and **React Js** and **Redux** for dynamic UI rendering achieving a 40% improvement in load times and increased user engagement across platforms.
- Architected and developed 10+ RESTful APIs using **Node.js** and **Express.js** for high-performance routing and asynchronous processing, resulting in a 30% reduction in backend latency and smoother data transaction flows.
- Enabled and maintained API security using **JWT** tokens for user authentication and authorization, and documented endpoints with Swagger to ensure clarity, and developer-friendly onboarding.
- Directed seamless integration of frontend components with backend APIs using **React**, **React hooks** and **Redux Toolkit**, leveraging **thunks** and **slices** for state management and asynchronous API handling, redesigned the UI to boost responsiveness by 35%, improving overall visual consistency.

Software Engineer, *Indiana University School of Education - Indianapolis*

06/2025 – Present

- Developed an LLM-powered NLP system to classify and summarize community-engaged research (CEnR) from unstructured academic documents, supporting automated content understanding.
- Built an end-to-end pipeline using **PyMuPDF** and **spaCy** for document parsing and preprocessing, and fine-tuned a **DistilBERT** model on 500 labeled samples to identify CEnR-relevant content.
- Integrated **GPT-4** with **Retrieval-Augmented Generation (RAG)** via **LangChain**, **Qdrant**, and **Sentence Transformers** to score engaged documents across 5 community research dimensions and deployed the system with a **Gradio** interface for real-time analysis and explainable outputs in **JSON/CSV**.

Software Engineer, *Societe Generale Global Solution Centre*

07/2022 – 08/2023 | Bangalore, India

MyFIN Gateway

- Implemented an AWS-hosted **API Gateway** to efficiently manage and route microservices-based REST and SOAP API requests across the CFT/FIN department, leveraging **Java** and **Spring Boot** for backend services to ensure scalable and secure integration and optimized request routing.
- Created and scheduled **AWS Lambda** functions written in Java to process XML and JSON data flowing between teams through Amazon Web Services API Gateway, resulting in a 40% reduction in data processing time and increased operational efficiency.
- Optimized cloud infrastructure by decommissioning unnecessary **EC2** server instances and consolidating **DynamoDB** databases and **AWS S3** buckets using **Terraform** resulting in a 15% reduction in overall cloud costs
- AWS tools such as **CloudWatch** were operated to monitoring, logging and Error Tracking, resulting in a 30% reduction in production delay.
- Worked with **AWS EventBridge**, **SNS** and **SQS** to automate event-driven workflows, reducing manual intervention by 80%.

Ballet

- Collaborated on the **UI and Java** development on the Ballet project, a fintech application that calculates employee bonuses in an Agile environment, ensuring continuous integration and delivery(CI/CD).
- Engineered interactive and user friendly dashboard with **React JS**, **React hooks** and **Redux** that visualizes core business KPIs from complex data, reducing the manual effort to report by 60%.
- Pioneered the development of scalable microservices and RESTful APIs using **Java** and **Spring Boot** for backend logic and service orchestration, **Spring Data JPA** and **Hibernate** for efficient ORM(object-relational mapping) and **Postgres** database access, **SQL** query tuning and achieving a 20% boost in system responsiveness.
- Authored end-to-end **Junit5** test suites and integration tests for various microservices contributing to enhanced code coverage and a 30% reduction in post-deployment issues.

PROJECTS

DeliverWise

09/2024 – 11/2024

- Developed a e-commerce Delivery Management System using **MERN** (MongoDB, Express JS, React JS, Node JS) stack.
- Established multi-factor authentication and social login within a modular microservices architecture, designing secure **RESTful API** endpoints for service selection, order tracking, registration, and quote management accelerating user onboarding by 25%.
- Built a user-friendly rating feature that improved customer transparency contributing to a 40% increase in user interaction and created dashboards with insights on delivery performance, customer satisfaction, and operational metrics reducing report generation time by 20%.
- Implemented **CI/CD** pipelines using **GitHub Actions** and containerized microservices with **Docker**, automating build, test, and deployment workflows to accelerate release cycles and ensure consistent development environments.

Social Network App

02/2024 – 05/2024

- Led the development of a student social networking application with real-time chat, enhancing peer-to-peer interaction and engagement.
- Delivered 7+ microservices API endpoints using **Java** and **Spring Boot**, integrating **OAuth 2.0** to enable secure and seamless user authentication.
- Crafted a responsive **Angular** frontend with modular components, reactive forms, and lazy-loaded routing, improving UI performance and ensuring smooth integration with backend services.
- Established **CI/CD** pipelines in **GitHub Actions** for automated deployments, increasing hosting efficiency and reducing release time by 40%.

LLM based Document Summarizer

05/2024 – 05/2024

- Engineered and prepared user-friendly **Streamlit** interface to enable document uploads and question-answer interactions.
- Customized **OpenAI GPT** model to generate document-aware, context-sensitive responses tailored for user queries.
- Built a **RAG** pipeline by fine-tuning HuggingFace's all-MiniLM-L6-v2 model for embedding generation, integrating **Chroma** vector database and **LangChain** for efficient retrieval and prompt orchestration, boosting response accuracy by 35%.

SKILLS

Languages — c++, Java, Python, SQL | **Databases** — MySQL , Postgres , NoSQL , MongoDB | **Frontend** — Javascript, Typescript, ReactJS, Angular | **Backend** — Springboot, REST APIs , NodeJs , MERN, Django, Flask API, FastAPI, AWS, Docker, Jenkins, Terraform, Maven |

Frameworks/Tools — LLMs, RAG, Numpy, Pandas, TensorFlow, Scikit-learn, Tableau, Power-BI, Jira, Eclipse, IntelliJ